AIDOOi

AID-DATA-005

PRODUCT IDENTIFICATION & VARIANT CODING STANDARD

1. Final Naming Architecture

AIDOOi uses non-semantic permanent identifiers. The identifier tells the system which record it is; taxonomy and attributes tell the AI what the product is.

Record	Format	Meaning
Parent Product	AIP-00000001	One underlying product model/family.
Product / Variant	AID-00000001	One canonical selectable physical variant.
Seller Offer	AIO-00000001	One seller-specific commercial offer.
Seller SKU	Seller supplied	Seller's external inventory/reference code.

2. Core Hierarchy

Main Category -> Category -> Product Type -> Parent Product -> Product / Variant

Parent Product is a relationship layer below taxonomy. Different product families can share exactly the same Product Type.

3. Multi-Category Example

Taxonomy / Family	Parent ID	Product IDs / Variants
Seating > Chair > Lounge Chair - Model A	AIP-00000001	AID-00000001 Brown AID-00000002 Beige AID-00000003 Black AID-00000004 Green
Tables > Living Room Table > Coffee Table - Model A	AIP-00000002	AID-00000005 Natural Oak AID-00000006 Dark Walnut AID-00000007 Black
Sofas > Sofa > Standard Sofa - Model A	AIP-00000003	AID-00000008 Beige Fabric AID-00000009 Grey Fabric AID-00000010 Green Fabric
Lighting > Ceiling Lighting > Pendant Light - Model A	AIP-00000004	AID-00000011 Brass AID-00000012 Black AID-00000013 Chrome AID-00000014 Bronze

4. Same Product Type, New Family

A second, fundamentally different Lounge Chair stays under the same taxonomy but receives a new Parent ID.

Taxonomy	Parent / Family	Products
Seating > Chair > Lounge Chair	AIP-00000001 - Model A	AID-00000001 Brown AID-00000002 Beige AID-00000003 Black AID-00000004 Green
Seating > Chair > Lounge Chair	AIP-00000005 - Model B	AID-00000015 Cream AID-00000016 Blue AID-00000017 Grey

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5. Variant Rule

Same underlying model + customer-selectable difference = same Parent.
Different fundamental model, structure or function = new Parent.

Difference	Same Parent?	Treatment
Colour	Yes	New AID Product linked to same AIP.
Upholstery / Fabric	Yes	New AID Product linked to same AIP.
Finish	Yes	New AID Product linked to same AIP.
Size	Conditional	Same Parent when officially sizes of the same model.
Orientation	Usually	Same Parent for left/right versions of same model.
Fundamental model/design change	No	Create a new AIP Parent.

6. Product With No Variants

A standalone product receives an AID Product ID and Parent Product ID = NULL. Artificial Parent records are not created.

7. Multi-Seller Example

Offer ID	Product ID	Seller	Seller SKU
AIO-00000001	AID-00000001	Blossom Home	RASO_D-6604
AIO-00000002	AID-00000001	Retailer B	CHAIR-BR-91
AIO-00000003	AID-00000001	Retailer C	LC-204-BR

8. Advantages Over Category-Embedded IDs

Area	New: AID-00000015	Old: AI-CHR-000015
Taxonomy independence	Excellent	Category meaning embedded in ID.
Reclassification	Taxonomy can change; ID stays fixed.	ID may retain outdated meaning.
AI matching	Uses explicit taxonomy and attributes.	Identifier may be mistaken for semantic data.
Scalability	New categories need no new ID convention.	New category abbreviations must be governed.
Consistency	One global Product ID pattern.	Multiple prefixes to maintain.
Human readability	Lower from ID alone.	Higher because category is visible.

9. AI Matching Rule

The Recommendation Engine must never infer product meaning from AID, AIP, AIO or Seller SKU. It must use explicit fields such as Main Category, Category, Product Type, dimensions, colour, material, style, price, stock and category-specific attributes.

10. Optional Human-Readable Classification Code

Permanent ID	Classification Code	Optional Display
AID-00000001	CHR	CHR | AID-00000001
AID-00000005	TBL	TBL | AID-00000005
AID-00000008	SOF	SOF | AID-00000008
AID-00000011	LGT	LGT | AID-00000011

11. Final Rule

ID = Which record? AID-00000015
Parent = Which family? AIP-00000005
Taxonomy = What is it? Seating > Chair > Lounge Chair
Attributes = What is it like? Blue, Fabric, Contemporary, dimensions, etc.
Offer = Who sells it and under what conditions? AIO-00000027

Supersession

This AID-DATA-005 replaces the previous version and supersedes the earlier category-embedded identifier concept such as AI-CHR-000001.